

CPEN 416 / EECE 5710

Lecture 15: Order finding and Shor's algorithm

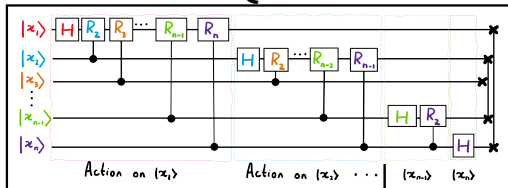
Thursday 30 October 2025

Announcements

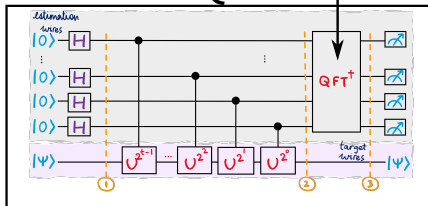
- Tutorial Assignment 3 due tomorrow 23:59
- Midterm checkpoint report due **tomorrow by 5:00pm**
 - Meeting appointments open in Canvas later this afternoon
- Tutorial 7 Monday: RSA and Shor's algorithm
- Quiz 7 Tuesday

Reminder: where are we going?

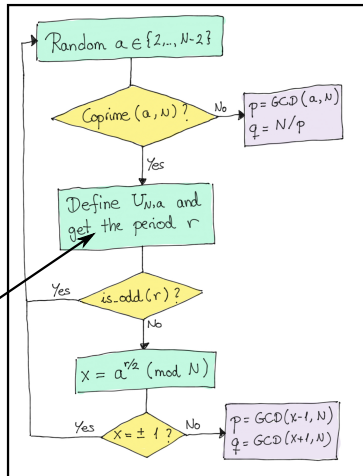
1. QFT



2. QPE



3. Shor



Last time

Input: integers a and N , where $N \geq 2$ and $\gcd(a, N) = 1$

Output: smallest integer r for which $a^r \equiv 1 \pmod{N}$

Exercise: find the order of $a = 8$ for $N = 15$.

Learning outcomes

- Apply QPE to solve the order finding problem
- Reduce order finding to integer factorization, and implement Shor's quantum algorithm in PennyLane
- Explain the relevance, and the societal and ethical implications, of Shor's algorithm

Order finding on a quantum computer

Given a, N with $\gcd(a, N) = 1$, consider the quantum operation for multiplication by a modulo N ,

If r is the order of a modulo N , and we apply $U_{N,a}$ r times,

Order finding on a quantum computer

$U_{N,a}$ has the same order, r . We can use QPE to efficiently find r with a clever initialization of the target register.

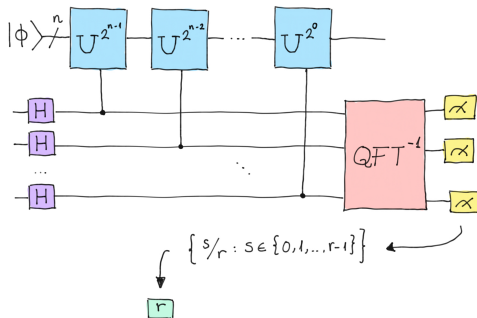


Image credit: Xanadu Quantum Codebook node S.3

Order finding on a quantum computer

Suppose there exists $|\phi\rangle$ for which

Consider the superposition state

Apply U to $|\Psi_0\rangle$:

Order finding on a quantum computer

Now consider the state

Apply U to $|\Psi_1\rangle$:

Order finding on a quantum computer

This generalizes to $|\Psi_s\rangle$

It has eigenvalue

Idea: create *any* $|\Psi_s\rangle$, run QPE to get estimate of s/r , compute r .

What's the problem?

Order finding on a quantum computer

Solution: construct the uniform superposition of all of them.

But what does this equal?

Order finding on a quantum computer

The superposition of all $|\psi_s\rangle$ is just our original state $|\phi\rangle$!

$$\begin{aligned}
 |\psi\rangle &= \frac{1}{\sqrt{r}} \left(|\psi_0\rangle + |\psi_1\rangle + \dots + |\psi_{r-1}\rangle \right) \\
 &= \frac{1}{\sqrt{r}} \left(\frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{\frac{-2\pi i}{r}} |\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)}{r}} |\phi\rangle \right) \right. \\
 &\quad \left. + \frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{\frac{-2\pi i(r-1)}{r}} |\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)^2}{r}} |\phi\rangle \right) \right. \\
 &\quad \left. + \dots + \frac{1}{\sqrt{r}} \left(|\phi\rangle + e^{\frac{-2\pi i(r-1)}{r}} |\phi\rangle + \dots + e^{\frac{-2\pi i(r-1)^{r-1}}{r}} |\phi\rangle \right) \right) \\
 &\quad \underbrace{\qquad\qquad\qquad}_{= \frac{1}{\sqrt{r}} \cdot \frac{1}{\sqrt{r}} \cdot r |\phi\rangle = |\phi\rangle}
 \end{aligned}$$

Diagram illustrating the simplification of the superposition of states $|\psi_s\rangle$ to the original state $|\phi\rangle$. The states are grouped into r terms, each containing r states. The phases cancel out, resulting in r copies of $|\phi\rangle$, which simplifies to $|\phi\rangle$.

If we run QPE, the output will be s/r for one of these states.

Image credit: Xanadu Quantum Codebook node S.3

Shor's algorithm: overview

Input: integer N , guaranteed to have structure

for p, q prime.

Output: p and q

Factoring is accomplished by a **reduction** from order finding.

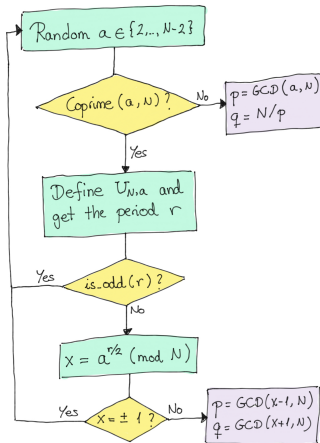


Image credit: Xanadu Quantum Codebook node S.3

Non-trivial square roots and factoring

Choose a random a that is coprime with N . If its order, r , is even,

Let's rewrite and factor...

Non-trivial square roots and factoring

This means

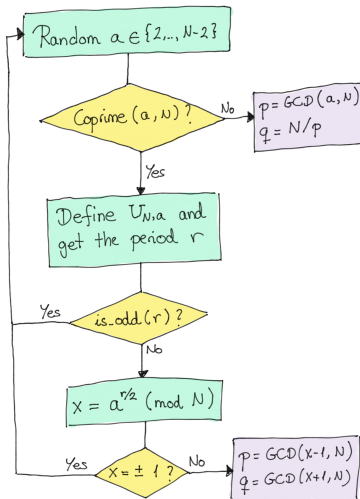
Based on structure of N ,

If $a^{r'} \neq \pm 1$, we can determine p and q

Is this really efficient?

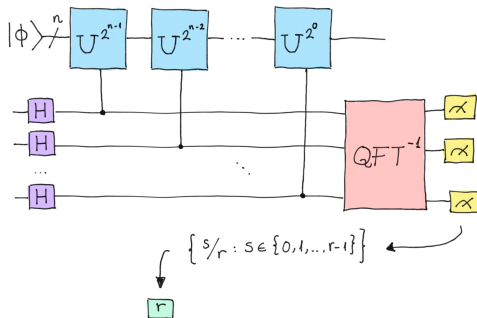
GCD: polynomial w/Euclid's algorithm

Modular exponentiation: can use exponentiation by squaring, other methods to reduce operations and memory required



Is this really efficient?

Quantum part: let $L = \lceil \log_2 N \rceil$.



QFT: gates polynomial in number of qubits $O(L^2)$

Controlled- U : requires $O(L^3)$ gates.

Discussion

Form pairs or small groups, and consider the following questions:

- 1 Shor's algorithm was developed in 1994. Estimate the fraction of today's world population that can actually implement it.
- 2 Shor's algo can be used to break cryptosystems like RSA. Estimate the proportion of the world that would be affected if someone actually deployed it at scale.
- 3 Is it ethical to develop such an algorithm? Is it ethical to *teach* such an algorithm?
- 4 Look up some resource estimates; how long would it actually take to break 2048-bit RSA? How many qubits are needed?
- 5 Think critically about (a) who knows how to implement the algorithm, and (b) who will potentially have access to quantum hardware in the future. What issues can you foresee?
- 6 What are ways we can keep our cryptographic infrastructure secure in the future?

Next time

Content:

- Tutorial 7: hands-on w/RSA and Shor's algorithm
- Module 4: quantum channels, and noise in quantum systems

Action items:

- ➊ Tutorial assignment 3
- ➋ Project midterm checkpoint report

Recommended reading:

- From this class: Codebook QFT, QPE, SH; N&C 5.3, A.5
- For next class: Codebook NT; N&C 8.1-8.3